

Pool Connection Guide

KV260 Bitcoin Mining - Connect to a Real Pool

Configure, build, and run the PDSA-FPGA miner on a live Stratum pool

Generated: 2026-06-02 22:57

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1. Prerequisites

Hardware

- Kria KV260 board with power adapter (12V, 3A)
- MicroSD card with Kria Ubuntu image (booted at least once)
- Ethernet cable connected to a router WITH internet access
- Laptop or PC on the same network (Wi-Fi or Ethernet)
- (Optional) Monitor + keyboard for first-time network setup

Software

- Windows: PowerShell, SCP client (built-in)
- Or: PuTTY (SSH), WinSCP (file transfer)
- Your PDSA-FPGA deployment folder with bitstreams and source code

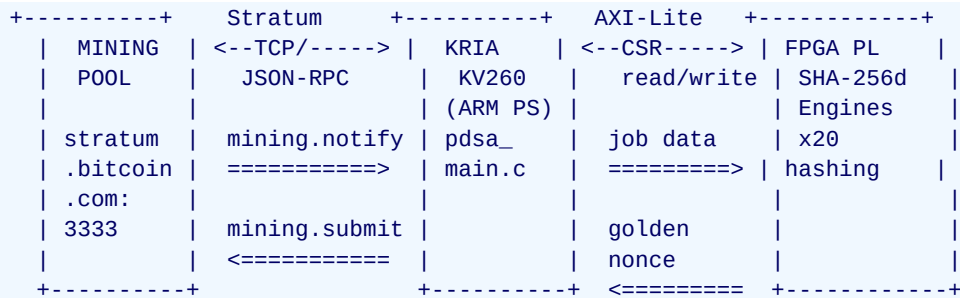
Pool Account

- A Bitcoin wallet address (e.g., bc1q... for Segwit)
- An account on a mining pool (free to register)
- A worker name (you create this in the pool dashboard)
- See Chapter 4 for pool setup instructions

WARNING: The KV260 MUST have internet access to connect to a mining pool. Direct Ethernet to laptop (no router) will NOT work for pool mining. Use the router connection.

2. How Pool Mining Works (Flow Diagram)

System Architecture



POOL sends work (block header template)
 KRIA distributes to 20 SHA-256d engines
 FPGA hashes at ~97 MHz, checks nonces
 When nonce found, submits back to pool
 Pool pays you when shares are accepted

The Stratum Protocol (v1)

The miner communicates with the pool using the Stratum protocol over TCP. Here is the conversation:

- 1. Connect: Open TCP socket to pool:3333
- 2. Subscribe: mining.subscribe -- tells pool who you are
- 3. Authorize: mining.authorize -- logs in your worker
- 4. Notify: mining.notify -- pool sends a new mining job
- 5. Submit: mining.submit -- you send back a found nonce
- 6. Repeat: pool sends new notify every ~30 seconds

TIP: The PDSA-FPGA miner handles ALL of the Stratum protocol automatically. You only need to configure the pool host, port, and worker name.

3. Step 1: Connect KV260 to the Internet

The KV260 must be on a network with internet access. The easiest way is to connect it to your home router via Ethernet.

Connection

1. Connect Ethernet cable from KV260 to your router (any LAN port)
2. Connect power adapter to KV260
3. Board powers on automatically
4. By default, KV260 uses DHCP, so it will automatically get an IP from your router
5. Wait ~30 seconds for boot to complete

Check Internet on KV260 (with Monitor)

If you have a monitor+keyboard connected:

```
ubuntu@kria:~$ ip addr show eth0  
Look for: inet 192.168.x.x/24
```

```
ubuntu@kria:~$ ping 8.8.8.8  
If you see replies, you have internet!
```

No Monitor? Use Phase 1 First

If you have not yet set up the KV260 with a monitor, follow the 'Direct Ethernet Guide' Phase 1 first: set a static IP, then later change to DHCP. Or connect a monitor briefly just to get the IP address.

WARNING: If you previously set a STATIC IP on the KV260 (for direct Ethernet to laptop), you must change it back to DHCP for pool mining. Otherwise the KV260 won't get internet.

4. Step 2: Find Your KV260's IP Address

You need the KV260's IP address to SSH into it. Here are several ways to find it.

Method 1: Check Router Admin Page

1. Open a browser on your laptop
2. Go to your router's admin page
(usually 192.168.0.1, 192.168.1.1, or 192.168.100.1)
3. Look for "DHCP Clients" or "Connected Devices"
4. Find a device named "kria" or "xilinx"
5. Note its IP address (e.g., 192.168.1.42)

Method 2: Use Angry IP Scanner (Windows)

1. Download Angry IP Scanner from angryip.org
2. Scan your subnet (e.g., 192.168.1.0-255)
3. Look for hostname "kria" or open port 22
4. The KV260 will show up as responsive

Method 3: Use nmap (PowerShell)

```
# Download nmap or use WSL:  
nmap -sn 192.168.1.0/24  
# Look for: "kria" or "xilinx"
```

Method 4: Monitor + Keyboard (Simplest)

```
Connect HDMI monitor and USB keyboard.  
Power on, login:  
Login:    ubuntu  
Password: ubuntu
```

```
Then run:  
ubuntu@kria:~$ hostname -I  
Output: 192.168.1.42
```

Write this IP down. You'll use it for SSH.

TIP: Once you know the IP, write it down. The IP may change if the router reboots (unless you set a DHCP reservation in the router).

5. Step 3: Transfer Deployment Files via SCP

Copy the deployment folder from your laptop to the KV260. Replace 192.168.1.42 with your KV260's actual IP.

On Your Laptop (PowerShell)

```
scp -r D:\pdsa_fpga_deployment  
ubuntu@192.168.1.42:~/pdsa_fpga_deployment
```

When prompted for password:
Type: ubuntu
(or your changed password)

Wait for transfer to complete (~30 seconds).

Verify on KV260 (via SSH)

```
ssh ubuntu@192.168.1.42  
Password: ubuntu  
  
ubuntu@kria:~$ ls ~/pdsa_fpga_deployment/  
bitstream/  
build_scripts/  
constraints/  
deployment/  
docs/  
ps_software/
```

If You Already Have Files

If you previously copied the files and only need to update the configuration, you can just edit the file in-place (see Step 5). No need to re-copy everything.

TIP: The password prompt doesn't show dots as you type. This is normal -- just type and press Enter.

6. Step 4: Create a Pool Account and Worker

You need a Bitcoin wallet address and an account on a mining pool. This guide covers the most popular pools.

Step 4a: Get a Bitcoin Wallet

Use any Bitcoin wallet that gives you a "Receive" address starting with "bc1q":

- Trust Wallet (phone app)
- Electrum (desktop)
- BlueWallet (phone)
- Exchange wallet (Binance, Coinbase)

Your wallet address looks like:

bc1qxy2kgdygjrsqtzq2n0yrf2493p83kkfjhx0wlh

This is where the pool sends your earnings.

Step 4b: Register on a Pool

Go to your chosen pool's website and create an account. Then create a worker:

Example: f2pool.com

1. Go to: <https://www.f2pool.com>
2. Click: Register (top right)
3. Enter email, password
4. Verify email
5. Login
6. Click: "Add Worker" or "Mining Account"
7. Enter Worker Name: "worker1"
8. Set Wallet Address: bc1qYOURWALLET...
9. Save

Your full worker for Stratum:

Username: bc1qYOURWALLET.worker1

Password: x (anything works, use "x")

Step 4c: Note Your Pool Connection Details

- Pool Host: e.g., btc.f2pool.com
- Pool Port: e.g., 3333
- Worker Username: bc1qYOURWALLET.worker1
- Worker Password: x (usually ignored)

TIP: You can use any pool. See Chapter 13 for a list of popular pools and their connection URLs.

7. Step 5: Configure Pool Settings in the Miner

Edit `pdsa_main.c` to set your pool details. You can do this on your laptop before transferring, or directly on the KV260 via SSH.

On KV260 (via SSH) - Edit the File

```
ssh ubuntu@192.168.1.42
Password: ubuntu

cd ~/pdsa_fpga_deployment/ps_software
nano pdsa_main.c
```

Find the Configuration Section

Look for lines 40-45. They contain the pool settings:

```
// Lines 40-45 in pdsa_main.c:
#define POOL_HOST      "pool.bitcoin.com"
#define POOL_PORT      3333
#define POOL_WORKER    "bc1qYOURWALLETADDRESS.worker1"
#define POOL_PASSWORD  "x"
#define POOL_API_URL    ""
```

Change to YOUR Pool

Replace the default values with your pool details from Step 4. Here is an example for f2pool:

```
#define POOL_HOST      "btc.f2pool.com"
#define POOL_PORT      3333
#define POOL_WORKER    "bc1qxyz123456789.worker1"
#define POOL_PASSWORD  "x"
#define POOL_API_URL    ""
```

How to Edit in nano

1. Use arrow keys to move to the line
2. Delete the old value between the quotes
3. Type your new value
4. Press: Ctrl+O (write out)
5. Press: Enter (confirm filename)
6. Press: Ctrl+X (exit)

To check your work:

```
grep "POOL_HOST|POOL_WORKER" pdsa_main.c
```

WARNING: Double-check your worker name format. Most pools use: YOURWALLET.WORKERNAME. The exact format depends on the pool. Check your pool's getting-started page.

8. Step 6: Load the FPGA Bitstream

Before running the miner, load the bitstream onto the FPGA. This configures the SHA-256d engines.

On KV260 (via SSH)

```
# Go to the deployment folder:
cd ~/pdsa_fpga_deployment

# Copy bitstream to firmware directory:
sudo cp bitstream/pdsa_full_sep_btc.bit
    /lib/firmware/

# Load via FPGA manager:
sudo sh -c '
    echo pdsa_full_sep_btc.bit >
    /sys/class/fpga_manager/fpga0/firmware'

# Verify it loaded:
cat /sys/class/fpga_manager/fpga0/state
# Should output: "operating"
```

If Using DFX with Separate Configs

If you built with the BD DFX separate flow (vivado_bd_dfx_separate_rp.tcl), you also have a partial bitstream for dynamic reconfiguration.

```
# Copy partial bitstream as well:
sudo cp bitstream/pdsa_rm_sep_btc.bit
    /lib/firmware/pdsa_rm_bitcoin.bit

# Or the .bin version if your system needs it:
sudo cp bitstream/pdsa_rm_sep_btc.bin
    /lib/firmware/pdsa_rm_bitcoin.bin
```

Check for Errors

```
dmesg | tail -10
# Should show:
# "fpga_manager: FPGA loaded successfully"
```

TIP: If 'cat state' shows 'operating', the FPGA is configured and ready. The blue LED on the KV260 should be steady (not blinking). Do this EVERY time you reboot the KV260.

9. Step 7: Install Build Tools and Build the Miner

The PS software (pdsa_main.c) must be compiled on the KV260's ARM processor.

Install Build Tools (First Time Only)

```
sudo apt update
sudo apt install build-essential make -y

This installs gcc, make, and related tools.
```

Build the Miner

```
cd ~/pdsa_fpga_deployment/ps_software

# Copy the Makefile if not already present:
cp ../deployment/Makefile ./

# Clean and build:
make clean
make -j4

# Verify the binary was created:
ls -la pdsa_miner
# Should show: -rwxr-xr-x ... pdsa_miner
```

If Build Fails

```
# Common fixes:

1. Missing build-essential:
sudo apt install build-essential make -y

2. Missing header files:
sudo apt install linux-headers-$(uname -r) -y

3. Check the Makefile contents:
cat ../deployment/Makefile

4. Try building with verbose output:
make clean; make V=1
```

TIP: You only need to rebuild the miner if you changed pdsa_main.c (e.g., pool configuration). If you just reboot, the binary is still there -- no need to rebuild.

10. Step 8: Run the Miner - Connected to Pool!

Now run the miner. It will connect to the pool and start hashing.

Run the Miner

```
cd ~/pdsa_fpga_deployment/ps_software

sudo ./pdsa_miner
```

What You Should See

The miner will first run self-tests, then connect to the pool. Here is the expected output:

```
=====
PDSA-FPGA Mining Controller v1.0
Kria KV260 - Bitcoin + Kaspas Adaptive
=====

[INIT] Mapping PL CSR registers...
[HAL] Sending job TEST0 to PL hardware (algo=Bitcoin)
[HAL] Job started. Target[0:3] = 00 00 00 00
[TEST] SHA-256 Test 1: PASS
[TEST] SHA-256 Test 2: PASS
[TEST] All hash tests PASSED
=====
ALL SELF-TESTS PASSED
=====

[POOL] Connecting to btc.f2pool.com:3333
       as bc1qYOURWALLET.worker1

[POOL] Subscribed (sid=abc123)
[POOL] Authorized (worker=bc1qYOURWALLET.worker1)
[HAL] Sending job JOB001 to PL hardware (algo=Bitcoin)
[HAL] Job started. Target[0:3] = 00 00 00 00
[STATUS] Algo=Bitcoin Hashrate=0 H/s
         Accepted=0 Rejected=0 DPR=0
[PDSA] PT=100% BCV=0.00 Decision=CONTINUE
```

Wait for the First Status Update

After 30 seconds, you will see the first real status update with hashrate. Give it a minute to warm up.

```
[STATUS] Algo=Bitcoin Hashrate=387000000 H/s
         Accepted=0 Rejected=0 DPR=0

Your hashrate should be approximately:
20 engines x ~97 MHz = ~388 MH/s (theoretical)
Real: ~300-387 MH/s

Don't worry if Accepted=0 for the first few
minutes. It takes time to find a valid share.
```

TIP: The miner runs as root (sudo). This is required for /dev/mem access to the FPGA registers. NEVER run the miner

without sudo.

11. Understanding the Output

Here is what each message type means:

Status Line (every 30 seconds)

```
[STATUS] Algo=Bitcoin Hashrate=387000000 H/s
         Accepted=10 Rejected=1 DPR=0
```

```
Algo:      Currently mining algorithm
           (Bitcoin or Kaspas)
Hashrate:  Current hash rate in H/s
           (divide by 1M for MH/s)
Accepted:  Shares accepted by pool
Rejected:  Shares rejected by pool
DPR:       Number of DPR switches so far
```

Nonce Found (rare)

```
[FOUND] *** NONCE FOUND! ***
         nonce=0x1A2B3C4D after 1234567890 hashes
```

This means one of your 20 engines found a valid nonce that meets the pool's target.

It is immediately submitted to the pool.
If accepted, Accepted counter goes up.

PDSA Evaluation (every 5 seconds)

```
[PDSA] PT=85% BCV=0.12 Decision=CONTINUE

PT:  Pool Token percentage (price/trading vol)
     High PT% = Bitcoin is profitable
     Low PT% = Consider switching to Kaspas
BCV:  Block Chain Value (network health)
Decision: CONTINUE = keep mining current algo
         SWITCH_PT = switch based on PT
         SWITCH_BCV = switch based on BCV
```

Pool Reconnection

```
[POOL] Disconnected - reconnecting in 10s...
```

The miner will automatically reconnect.
This is normal if the pool restarts or there is a brief network interruption.

Stopping the Miner

Press: Ctrl+C

```
[MAIN] Shutting down...
[MAIN] Stopping engines...
[MAIN] Final stats:
```

```
Total hashes:      1234567890
Shares accepted: 10
Shares rejected: 1
DPR switches:      0
[MAIN] Log saved to /tmp/pdsa_measurements.csv
```

Measurement Log

The miner saves detailed statistics to `/tmp/pdsa_measurements.csv`. This file contains one row per status interval with: timestamp, algorithm, hashrate, accepted, rejected, pool token, BCV, decision. You can analyze this later to evaluate performance.

TIP: It is normal to NOT see '[FOUND]' messages for many minutes. The Bitcoin network difficulty is very high. An FPGA at 388 MH/s will find shares occasionally, but not frequently. This is expected for hobbyist mining.

12. Step 9: Run Persistently (screen / systemd)

If you close your SSH session, the miner stops. Use one of these methods to keep it running.

Method 1: screen (Simple)

```
# Install screen (first time only):
sudo apt install screen -y

# Start a screen session:
screen -S miner

# Run the miner:
cd ~/pdsa_fpga_deployment/ps_software
sudo ./pdsa_miner

# Detach from screen:
Press: Ctrl+A, then D

# You are back at the shell.
# The miner keeps running.

# To re-attach and see output:
screen -r miner

# To list all screen sessions:
screen -ls
```

Method 2: nohup (Simple)

```
# Run in background:
cd ~/pdsa_fpga_deployment/ps_software
nohup sudo ./pdsa_miner &

# Output goes to nohup.out
# To check output:
tail -f nohup.out

# To stop:
sudo pkill pdsa_miner
```

Method 3: systemd Service (Auto-start on Boot)

```
# Create a service file:
sudo nano /etc/systemd/system/pdsa-miner.service
```

Service File Contents

```
[Unit]
Description=PDSA-FPGA Mining Controller
After=network.target multi-user.target

[Service]
Type=simple
User=root
WorkingDirectory=/home/ubuntu/
pdsa_fpga_deployment/ps_software
```



```
ExecStartPre=/bin/sh -c 'echo pdsa_full_sep_btc.bit >
    /sys/class/fpga_manager/fpga0/firmware'
ExecStart=/home/ubuntu/
    pdsa_fpga_deployment/ps_software/pdsa_miner
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
```

```
# Enable and start the service:
sudo systemctl daemon-reload
sudo systemctl enable pdsa-miner
sudo systemctl start pdsa-miner

# Check status:
sudo systemctl status pdsa-miner

# View logs:
sudo journalctl -u pdsa-miner -f
```

WARNING: Method 3 (systemd) is best for production. The miner will automatically start when the KV260 boots, restart if it crashes, and log to journalctl. Use this if you plan to run 24/7.

13. Pool Examples

Here are the connection details for popular Bitcoin mining pools. Replace 'YOURWALLET' with your actual wallet address.

f2pool (Recommended)

```
Pool Host:  btc.f2pool.com
Port:       3333 (or 3334 for backup)
Worker:     YOURWALLETADDRESS.worker1
Password:   x

Website:    https://www.f2pool.com
Minimum payout: 0.001 BTC
Fee: 2.5%
```

ckpool (Solo Mining)

```
Pool Host:  solo.ckpool.org
Port:       3333
Worker:     YOURWALLETADDRESS.worker1
Password:   x

Website:    https://solo.ckpool.org
Note: SOLO mining. You get the ENTIRE
block reward if you find a block, but
it's much rarer than pooled mining.
```

ViaBTC

```
Pool Host:  btc.viabtc.com
Port:       3333
Worker:     YOURWALLETADDRESS.worker1
Password:   x

Website:    https://www.viabtc.com
Fee: 1%
```

Poolin (Shutting Down)

```
Pool Host:  btc.poolin.com
Port:       3333
Worker:     YOURWALLETADDRESS.worker1
Password:   x

Website:    https://www.poolin.com
Note: Poolin winding down operations.
Use f2pool or ViaBTC instead.
```

Public Test Pool (For Testing)

If you just want to test connection
without a real wallet:

```
Pool Host:  stratum.bitcoin.org
Port:       3333
Worker:     test.test
Password:   x
```

May not always be available.

How to Switch Pools

To switch pools, edit pdsa_main.c and change the POOL_HOST and POOL_WORKER lines, rebuild with 'make clean; make -j4', then restart the miner.

TIP: Start with f2pool or ViaBTC. They are reliable, easy to set up, and have good support. Solo mining (ckpool) is not recommended for 388 MH/s - you may never find a block.

14. Troubleshooting

Cannot SSH: Connection Refused

```
# SSH server not running on KV260?
# You need a monitor+keyboard to fix this:

# On KV260:
sudo systemctl status ssh
sudo systemctl enable --now ssh

# If ssh not installed:
sudo apt install openssh-server -y
```

Cannot SSH: Connection Timed Out

```
# Wrong IP address?
# Check the IP again (see Step 2)

# KV260 not powered on?
# Check green LED

# Different subnet?
# Your laptop and KV260 must be on the
# SAME network (e.g., both 192.168.1.x)

# Try pinging the KV260 first:
ping 192.168.1.42
```

Miner: '[POOL] Connection failed'

```
# Check internet on KV260:
ping 8.8.8.8

# Check DNS resolution:
nslookup btc.f2pool.com

# Check pool hostname/port in pdsa_main.c:
grep "POOL_HOST\|POOL_PORT" pdsa_main.c

# Test TCP connection:
timeout 5 bash -c
'echo > /dev/tcp/btc.f2pool.com/3333'
# If no output, port is reachable
```

Miner: '[POOL] Handshake failed'

```
# Wrong worker name format?
# Some pools use: username.worker
# Others use: username_worker
# Check your pool's documentation

# Wrong wallet address?
# Double-check for typos
```

```
# Try Stratum test:
# (On another machine with Python)
# python -c "
# import socket
# s=socket.socket()
# s.connect(('btc.f2pool.com', 3333))
# s.send(b'{"id":1,"method":
#         "mining.subscribe",
#         "params":[]}\n')
# print(s.recv(1024))
# "
```

Miner: '[INIT] HAL init failed'

```
# Forgot sudo?
sudo ./pdsa_miner
# (not: ./pdsa_miner)

# Bitstream not loaded?
cat /sys/class/fpga_manager/fpga0/state
# Should be "operating"

# Wrong device tree?
# Check /dev/mem exists:
ls -la /dev/mem
```

FPGA Bitstream Load Fails

```
# Check state:
cat /sys/class/fpga_manager/fpga0/state

# Check if file exists:
ls -la /lib/firmware/pdsa_full_sep_btc.bit

# Check kernel messages:
dmesg | tail -20

# Try re-copying:
sudo cp ~/pdsa_fpga_deployment/bitstream/
pdsa_full_sep_btc.bit /lib/firmware/
```

Hashrate is 0 H/s

```
# Bitstream loaded? Check fpga state.
# Miner running with sudo?
# Wait for first status update (30s).

# Check FPGA clocks:
cat /sys/kernel/debug/clk/clk_summary
# Look for PL clock (should be ~97 MHz)

# Check CSR register access:
sudo ./pdsa_miner --verbose
# (if verbose mode exists)
```

Many Shares Rejected

```
# High rejection rate (>10%) indicates:
# - Poor timing closure (unlikely, slack=0.6ns)
# - Incorrect nonce range
# - Pool network latency

# Check in pdsa_main.c if the nonce
# range is configured correctly.

# Try a pool closer to your region.
# f2pool has servers in US, EU, Asia.
```

KV260 Overheating / Unstable

```
# The FPGA draws significant power.
# Ensure adequate cooling:
# - Fan on the KV260 heatsink
# - Airflow around the board
# - Ambient temperature < 35C

# Check temperature:
cat /sys/class/thermal/thermal_zone*/temp
# Divide by 1000 to get Celsius

# If > 85C, improve cooling.
# The KV260 will throttle at ~100C.
```

Can't Find KV260 on Network

```
# Ethernet cable plugged in?
# Router has DHCP enabled?
# Try a different LAN port on router.

# If you changed to static IP (for direct
# Ethernet), change back to DHCP:
sudo nano /etc/netplan/01-netcfg.yaml
# Set: dhcp4: true
sudo netplan apply

# If all else fails, use monitor+keyboard:
hostname -I
```

Forgot SSH Password

```
# Default password: "ubuntu"
# If you changed it and forgot,
# you need a monitor+keyboard:

# Boot with monitor, login, reset:
passwd ubuntu

# Or use the Kria recovery image
# to re-flash the SD card.
```